ECOPRISM

Sustainability as a Competitive Advantage

From Reporting to Strategic Decision-Making

Executive summary

Sustainability has rapidly evolved from a **voluntary corporate responsibility** initiative into a **strategic priority** shaped by regulation, capital markets, and changing consumer expectations. Companies now face growing pressure to disclose environmental and social impacts, manage climate risks, and demonstrate responsible supply chain practices. However, despite increasing attention and investment, many organizations struggle to translate ESG commitments into measurable competitive advantage.

A key reason is that ESG is often implemented as a reporting or compliance function rather than as a strategic capability. In many organizations, sustainability responsibilities remain fragmented across departments such as **compliance, investor relations, procurement, and communications.** As a result, ESG initiatives frequently improve disclosure and reputation but remain disconnected from the decisions that shape enterprise performance including capital allocation, product innovation, operational strategy, and executive incentives. This creates a structural gap between sustainability ambition and value creation.

This whitepaper argues that ESG becomes a source of competitive advantage when embedded directly into strategic business decisions. The pathways to competitive advantage vary by sector ranging from cost efficiency in **manufacturing,** to brand differentiation in **FMCG**, and improved project viability in **infrastructure.**

To bridge this gap, the paper introduces a **Five-Step Strategic ESG Integration Framework** - a practical pathway for embedding sustainability into core business processes, from identifying material issues to measuring financial outcomes.

Ultimately, companies that treat sustainability as a strategic capability rather than a compliance exercise will be better positioned to drive growth, manage risks, and sustain **long-term competitive advantage.**

Why ESG efforts fail to deliver strategic advantage

Over the past decade, ESG has moved from **voluntary corporate responsibility** to **a board-level priority** shaped by regulation, capital markets, and customer expectations. Sustainability disclosures are expanding, climate risk is entering financial filings, and supply-chain transparency is becoming a commercial requirement rather than a reputational choice. Yet despite this momentum, many organizations struggle to convert ESG ambition into measurable strategic advantage.

Global capital markets are **increasingly pricing sustainability performance**, with sustainable debt markets (including green, social, and sustainability-linked bonds) surpassing **USD 4 trillion cumulatively**, indicating growing integration of ESG factors into financing structures.[1]

Regulatory expectations are intensifying, with frameworks like the **EU's CSRD** and the **California Climate Corporate Data Accountability Act** requiring detailed emissions disclosures, shifting ESG from narrative reporting to auditable, decision-grade data.

Despite this **convergence of regulatory, investor, and consumer pressure**, ESG often fails to produce competitive differentiation. In various cases:

- Sustainability targets are published but not embedded into capital allocation
- Emissions are measured but not linked to cost optimization
- Supplier codes of conduct exist, but procurement incentives remain unchanged
- ESG reports improve in volume, but strategic decision-making remains financially siloed

The result is a structural paradox: **ESG visibility is increasing, but ESG-driven value creation remains inconsistent.**

Research by McKinsey & Company shows that while over **90% of executives consider ESG important**, far fewer integrate it systematically into core business strategy and investment decisions.[2]

This gap between ambition and integration explains why **ESG initiatives frequently underperform strategically.** The challenge is not the absence of frameworks or disclosures; it is the absence of structural alignment between sustainability objectives and enterprise value creation.

Two underlying causes consistently emerge:

1. Fragmented governance and decision-making

One of the most persistent barriers to ESG-driven competitive advantage is fragmented governance. In many organizations, sustainability responsibilities are distributed across compliance, risk, procurement, investor relations, operations, and corporate communications often without a single point of accountability or decision-making authority.

1. World Investment Report 2024: Chapter III - Sustainable finance trends
2. ESG impact and effective ESG operating models | McKinsey

This structural fragmentation creates three systemic weaknesses:

a) ESG without enterprise ownership

Across many organizations, ESG initiatives involve multiple functions, yet clear operational leadership is often lacking. While boards increasingly discuss ESG priorities, responsibility for translating these commitments into day-to-day business decisions, performance metrics, and investment choices frequently remains unclear.

A survey reported by CFO Dive found that **only 9% of finance departments have primary oversight of ESG,** while responsibility is spread across sustainability (27%), investor relations (24%), legal teams (23%), and others (17%). This fragmentation highlights the absence of a centralized leadership structure capable of integrating ESG into financial and operational decision-making.[3]

Without clear ownership tied to financial authority, ESG remains advisory rather than directive, limiting its influence on investment decisions, procurement policies, and risk management frameworks.

b) Disconnected data systems and methodologies

Fragmented governance is often reinforced by fragmented data. ESG information is frequently collected through spreadsheets, manual surveys, and disconnected systems across functions.

A 2023 survey by Deloitte found that **57% of executives identify data quality and fragmentation as their top ESG challenge,** constraining decision-making and assurance readiness.[4]

As frameworks like the Corporate Sustainability Reporting Directive increase assurance requirements, inconsistent methodologies and undocumented assumptions raise compliance and reputational risks. Without centralized systems, ESG cannot serve as a reliable management tool.

c) Weak linkage to executive incentives

Competitive advantage requires aligned incentives. While adoption has increased, ESG incentives remain limited relative to financial metrics.

Research shows that the integration of ESG metrics into executive pay is still uneven. The share of companies linking executive compensation to ESG performance increased from just **1% in 2011 to about 38% in 2021.**[5]

When executive rewards remain tied primarily to short-term financial performance, sustainability objectives compete with earnings targets, reducing the likelihood of long-term ESG-driven transformation.

3. Just 9% of finance departments have primary ESG oversight: survey | CFO Dive
4. 2024-sustainability-action-report.pdf
5. ESG Performance Metrics in Executive Pay

2. Misalignment between ESG and core business strategy

Even when governance structures exist, ESG frequently fails because it is misaligned with the company's economic engine. Instead of being integrated into revenue generation, cost structures, and capital efficiency models, sustainability initiatives are treated as compliance obligations or reputational safeguards.

This misalignment manifests in three critical ways:

a) Capital allocation disconnect

Many companies announce net-zero or sustainability commitments but fail to embed them into capital allocation decisions, which often continue to rely on traditional financial metrics without accounting for carbon pricing, transition risks, or long-term climate scenarios.

An EY survey found that **72% of global CFOs believe their capital allocation processes need improvement**, while only **40% say their capital allocation approach is flexible enough** to respond to evolving environmental and socio-economic challenges.[6]

b) Failure to connect ESG to revenue and growth

Sustainability increasingly influences both investor and consumer behavior, yet many organizations do not translate ESG capabilities into commercial differentiation.

A global survey by Nielsen found that **73% of global consumers say they would change their consumption habits to reduce environmental impact**, indicating strong demand for more sustainable products and services.[7]

c) ESG as a cost center rather than a value driver

In many organizations, ESG is still treated primarily as a compliance obligation focused on reporting and regulation rather than as a driver of financial performance, causing sustainability initiatives to compete with revenue-generating investments for budgets.

Yet most companies still treat sustainability as a reporting obligation rather than a growth lever. Organizations that embed ESG into product innovation, pricing strategy, and market positioning are better placed to capture this demand, and convert consumer intent into measurable revenue.

Companies focusing on material sustainability factors have been shown to outperform peers by **4–6% annually in stock returns**, driven by integrating ESG into growth strategies and shareholder value creation.[8]

The core challenge is not ESG's relevance - it's the structural gap between sustainability ambition and strategic execution.

6. Using Capital Allocation to drive long-term value. | EY - South Africa
7. Achieving sustainable profitable growth with ESG | McKinsey

The Business Case: How ESG drives competitive advantage

As ESG expectations intensify across regulators, investors, and customers, the strategic question for businesses is no longer whether sustainability matters, but how it can create measurable competitive advantage. Regulatory frameworks, disclosure standards, and stakeholder scrutiny are increasingly shaping corporate priorities, requiring organizations to move beyond compliance and reporting.

Organizations that integrate sustainability into core strategy are increasingly demonstrating that ESG can generate tangible business value. Companies that integrate ESG into core decision-making such as product design, supply chain management, capital allocation, and operational processes are increasingly demonstrating advantages in revenue growth, cost efficiency, and long-term resilience.

1. Revenue growth and market access

Sustainability is increasingly influencing consumer purchasing behavior, supplier selection, and access to international markets. Firms that incorporate ESG considerations into product design, sourcing practices, and operational transparency are better positioned to capture new revenue opportunities and maintain access to global value chains.

For many companies, this shift requires integrating ESG considerations directly into strategic decision-making. Product development, pricing strategies, and market entry decisions increasingly depend on environmental and social performance. Embedding sustainability criteria into these core business decisions enables firms to translate ESG commitments into differentiated products, stronger customer trust, and sustained market growth.

Research from the NYU Stern Center for Sustainable Business shows that sustainability-marketed products accounted for **54.7% of growth in the consumer-packaged goods market between 2015 and 2019,** despite representing a **16.1%** share of total products.[8]

Beyond consumer markets, ESG performance is becoming a prerequisite for participating in global supply chains. Many multinational buyers now evaluate suppliers based on environmental and social performance metrics, including emissions disclosures and responsible sourcing practices. Companies that demonstrate credible ESG performance are therefore better positioned to secure long-term procurement partnerships.

Sustainability also affects export competitiveness. Policies such as the European Union's Carbon Border Adjustment Mechanism introduce carbon pricing for imports of emissions-intensive products. Manufacturers with lower carbon footprints will be better positioned to maintain access to European markets and avoid additional compliance costs.

Schneider Electric: Sustainability-led growth

A strong example of sustainability driving revenue growth is Schneider Electric. The company has embedded sustainability into its core product portfolio, including energy-efficient infrastructure, digital energy management solutions, and decarbonization services. As demand for low-carbon solutions has increased, Schneider Electric has positioned sustainability as a central component of its business strategy.

According to the company's sustainability disclosures, **over 70% of Schneider Electric's annual revenue is now linked to products and solutions that contribute to energy efficiency and sustainability outcomes.**[9]

8. Sustainable Market Share Index™ - NYU Stern
9. ESG strategy FAQ | Schneider Electric

2. Cost optimization & operational efficiency

Many sustainability initiatives generate direct financial benefits by improving energy efficiency, reducing material waste, and optimizing resource use across operations. When embedded into operational strategy, ESG initiatives can function as powerful drivers of productivity and cost competitiveness.

Achieving these efficiency gains often requires integrating ESG considerations directly into operational decision-making. Investments in energy-efficient equipment, circular resource management, and sustainable production processes increasingly form part of strategic capital allocation and operational planning.

Companies that incorporate sustainability criteria into these decisions are better positioned to reduce operating costs, improve resource productivity, and build long-term operational competitiveness.

According to the International Energy Agency, the industrial sector accounts for **around 37% of global energy consumption**, highlighting the significant cost savings potential associated with improved energy productivity.[10]

Companies are increasingly adopting resource-efficient production processes, waste-to-value models, and circular resource management practices. These approaches not only reduce environmental impact but also improve operational resilience by lowering exposure to volatile energy and material prices.

Unilever – Operational resource efficiency

Unilever has integrated sustainability initiatives into its manufacturing operations to improve resource efficiency and reduce operational costs. Through its **Sustainable Living Plan**, the company implemented energy efficiency programs, waste reduction strategies, and water optimization measures across its global production facilities.

Between 2008 and 2020, Unilever reduced **CO_2 emissions from manufacturing by over 65% while cutting energy costs per ton of production by more than 40%.** The company also achieved **zero non-hazardous waste to landfill across its global factory network,** reducing waste disposal costs and improving production efficiency.[11]

This example demonstrates how embedding sustainability into operational processes can simultaneously **reduce costs, improve resource productivity, and strengthen long-term operational resilience.**

10. Industry - Energy System – IEA
11. Unilever Climate Transition Action Plan updated 2024

3. Risk mitigation & supply chain resilience

Integrating ESG into strategic decision-making also strengthens an organization's ability to anticipate and manage environmental, regulatory, and operational risks. As climate change, resource scarcity, and regulatory scrutiny intensify, companies are increasingly exposed to disruptions across their operations and supply chains. Organizations that proactively manage these exposures through ESG frameworks are better positioned to maintain operational continuity and safeguard long-term value.

Embedding ESG considerations into enterprise risk management and supply chain strategy enables organizations to move from reactive compliance to proactive resilience planning. Strategic decisions related to **supplier selection, geographic diversification, infrastructure investment, and climate adaptation** increasingly depend on understanding environmental and social risks across the value chain.

Companies that integrate these insights into long-term planning can reduce disruption risks, protect critical operations, and strengthen their competitive position in an increasingly volatile global business environment.

Climate-related risks represent one of the most significant emerging business challenges, with **potential financial impacts exceeding $1 trillion globally**, highlighting the scale of exposure businesses face from climate change and transition risks.[12]

Supply chain risks are also becoming increasingly prominent. Many companies are discovering that most of their environmental and social impacts lie outside their direct operations. CDP reports **that supply chain emissions are on average more than 11 times higher than companies' operational emissions**, highlighting the importance of monitoring and managing value-chain risks.[13]

12. Climate-related risks and opportunities
13. Climate Change Starts with Supply Chains | BCG

Sector deep-dive

Turning ESG into sector-specific moats

While ESG principles are broadly applicable across industries, the mechanisms through which sustainability creates competitive advantage vary significantly by sector. The most successful organizations do not apply ESG as a generic framework; instead, they translate sustainability priorities into **sector-specific operational, regulatory, and market advantages.**

In practice, ESG becomes strategically valuable when it strengthens a company's ability to:

- Reduce operational costs
- Manage regulatory and supply-chain risks
- Access international markets and capital
- Differentiate products and services

The following sector perspectives illustrate how ESG can evolve from a compliance requirement into a **defensible competitive moat**.

Manufacturing: ESG as operational efficiency & export shield

Manufacturing is one of the most resource-intensive sectors, with high energy consumption, material inputs, and complex global supply chains. As a result, ESG performance increasingly affects **operational costs, regulatory exposure, and access to international markets.** Companies that integrate sustainability into production processes and supply chain management are able to improve efficiency while strengthening competitiveness in global value chains.

Key ESG levers

1. Energy efficiency and decarbonization
Energy represents a major operating cost in manufacturing. Investments in energy-efficient technologies, electrification of industrial processes, and renewable energy adoption can significantly reduce both emissions and operational expenses.

The **World Economic Forum** estimates that **digital technologies and advanced efficiency solutions could reduce industrial emissions by up to 20% by 2050 while improving productivity.**[15]

15. Digital technologies can cut global emissions by 20%. Here's how | World Economic Forum
16. Making the $4.5 trillion circular economy opportunity a reality | World Economic Forum

Improving energy productivity therefore offers manufacturers a direct pathway to lowering operational costs while preparing for tightening climate regulations. Over time, companies that reduce energy intensity are better positioned to manage energy price volatility, comply with emerging carbon policies, and strengthen their competitiveness in global markets increasingly prioritizing low-carbon production.

2. Resource and waste optimization
Circular production models and material efficiency initiatives allow manufacturers to reduce raw material usage and production waste. These practices not only lower input costs but also reduce environmental impact across manufacturing operations.

Research from World Economic Forum estimates that **circular economy practices in manufacturing could generate $4.5 trillion in economic benefits globally by 2030.**[16]

By improving material productivity, companies can strengthen cost efficiency while reducing exposure to resource price volatility. Organizations that embed circular production practices into their operations also gain long-term resilience by minimizing input dependencies and positioning themselves as sustainable suppliers within increasingly sustainability-driven global value chains.

3. Low-carbon supply chain integration
Manufacturers are increasingly required to track environmental and social impacts across suppliers, particularly for Scope 3 emissions. As global buyers and regulators demand greater transparency, supplier-level sustainability data has become a critical component of ESG strategy.

Scope 3 emissions can account for around 75% of a company's total greenhouse-gas footprint on average, highlighting the critical role of value-chain transparency and supplier engagement in corporate decarbonization strategies.[17]

Improved supply chain transparency enables manufacturers to meet regulatory disclosure requirements and maintain eligibility in global procurement networks. Companies that proactively engage suppliers on emissions and sustainability standards can also strengthen supply chain resilience, reduce reputational risks, and secure long-term partnerships with multinational buyers prioritizing responsible sourcing.

ABB – ESG in industrial sales strategy

ABB sought to integrate environmental impact insights directly into its service quotations, enabling customers to evaluate the sustainability implications of different product & service. However, calculating environmental impacts manually was complex and time-consuming, making it difficult to present consistent sustainability metrics during the sales process.

Using ecoQUOTE, ABB integrated automated environmental impact calculations into it's **configure-price-quote (CPQ) system**, allowing emissions and material impacts to be calculated and embedded directly into customer quotations. This enabled sales teams to communicate the environmental benefits of products & service options clearly and consistently in real time.

By embedding sustainability metrics into commercial workflows, ABB strengthened transparency with customers, aligning its services with growing demand for lower-impact industrial solutions.

15. Digital technologies can cut global emissions by 20%. Here's how | World Economic Forum
16. Making the $4.5 trillion circular economy opportunity a reality | World Economic Forum

FMCG: Sustainability as brand and supply chain power

In the fast-moving consumer goods (FMCG) sector, sustainability is increasingly shaping **brand perception, consumer loyalty, and supply chain resilience**. Unlike heavy industries where ESG advantages often arise from operational efficiencies, FMCG companies derive strategic value from sustainability through **consumer trust, responsible sourcing, and transparent supply chains.**

As consumers, retailers, and regulators place greater emphasis on environmental and social responsibility, sustainability is becoming a key differentiator influencing purchasing decisions and long-term brand equity.

In increasingly competitive consumer markets, sustainability is increasingly shaping how companies compete for shelf space, retailer partnerships, and market share. Retailers and global brands are placing stricter sustainability requirements on suppliers, while consumers are gravitating toward products that align with environmental and social values.

As a result, companies that proactively embed sustainability into sourcing, packaging, and brand strategy are better positioned to secure distribution channels, strengthen retailer relationships, and capture long-term market growth.

Key ESG levers

1. Sustainable sourcing

FMCG companies rely heavily on agricultural commodities such as palm oil, cocoa, coffee, and soy, which are often linked to environmental and social risks including deforestation and labor practices. As a result, responsible sourcing and supplier traceability have become critical components of ESG strategy.

According to the Food and Agriculture Organization, **agrifood systems account for roughly one-third of global greenhouse gas emissions**, much of which originates within agricultural supply chains.[18]

Strengthening sustainable sourcing practices enables FMCG companies to mitigate environmental risks, maintain compliance with emerging supply chain regulations, and reinforce brand credibility among increasingly sustainability-conscious consumers.

17. How scope 3 emissions are key to decarbonization | World Economic Forum

2. Packaging innovation
Packaging represents one of the most visible sustainability challenges for FMCG companies. Plastic waste, recyclability, and circular packaging models have become major areas of focus for both regulators and consumers.

The United Nations Environment Programme estimates that **over 400 million tonnes of plastic are produced globally each year,** with packaging accounting for the largest share.[19]

Companies that invest in recyclable, reusable, or biodegradable packaging can reduce regulatory risk while strengthening brand reputation and responding to growing demand for environmentally responsible products.

3. Consumer trust and sustainable brand positioning
Consumers increasingly expect clear information about product sustainability, including sourcing practices, environmental impact, and ethical standards. Transparent labeling and credible sustainability claims are therefore becoming essential components of FMCG strategy.

A survey by PwC found that **consumers are willing to pay an average premium of nearly 10% for sustainably produced goods**, highlighting the growing influence of sustainability on purchasing behavior.[20]

Companies that communicate sustainability commitments effectively and integrate them into product development and branding strategies are better positioned to capture consumer preference and build long-term brand loyalty.

CASE STUDY

Nestlé – Sustainable sourcing as a competitive advantage

Nestlé has integrated sustainability into its sourcing and supply chain strategy to strengthen long-term competitiveness in the global food and beverage market. Through initiatives such as responsible sourcing programs, supplier traceability systems, and investments in regenerative agriculture, the company has embedded ESG considerations into key operational and procurement decisions.

A central pillar of this strategy is Nestlé's focus on **deforestation-free and fully traceable supply chains**, particularly in high-risk commodities such as cocoa and coffee. By improving transparency and working directly with farmers, Nestlé ensures compliance with emerging global regulations while reducing supply chain risks.

This strategy has enabled Nestlé to strengthen supplier reliability, maintain access to global retail markets, and align with increasing regulatory and consumer expectations around ethical sourcing.

18. Food systems account for more than one third of global greenhouse gas emissions
19. Taking on plastic pollution | UNEP Annual Report

Infrastructure & industrial services: ESG as license to operate

Infrastructure and industrial service providers operate in highly regulated and capital-intensive environments where environmental and social impacts are closely scrutinized. Projects often involve long development cycles, significant land use, and multiple stakeholders, including regulators, local communities, and institutional investors.

As a result, ESG performance increasingly determines whether projects receive approvals, attract financing, and maintain long-term operational stability. In this sector, sustainability is not only a compliance requirement but a critical factor in securing the **"license to operate."**

In infrastructure and industrial services, ESG performance increasingly influences strategic project decisions, including site selection, financing structures, and long-term asset development. Companies that integrate environmental and social considerations into these decisions are better positioned to secure regulatory approvals, attract institutional investors, and deliver large-scale projects with fewer delays and disputes.

Key ESG levers

1. Environmental impact management
Infrastructure projects must carefully manage environmental impacts such as emissions, land use, biodiversity loss, and water consumption. These factors are increasingly subject to regulatory scrutiny and environmental impact assessments.

According to the United Nations Environment Programme, the **buildings and construction sector accounts for around 37% of global energy-related CO_2 emissions,** highlighting the significant environmental footprint of infrastructure development.[21]

Strengthening environmental impact management enables companies to reduce regulatory risks, accelerate project approvals, and align infrastructure investments with global decarbonization goals.

2. Project delivery and stakeholder alignment
Large infrastructure projects involve multiple stakeholders, including governments, regulators, local communities, and investors. Misalignment between these stakeholders can lead to project delays, legal disputes, and rising development costs.

Research supported by the Inter-American Development Bank found that **community concerns were a contributing factor in about 70% of infrastructure project conflicts,** demonstrating how stakeholder misalignment can disrupt large-scale infrastructure development.[22]

20. Consumers willing to pay 9.7% sustainability premium, even as cost-of-living and inflationary concerns weigh: PwC 2024 Voice of the Consumer Survey | PwC

Infrastructure developers that integrate stakeholder considerations into early project planning and decision-making are better positioned to manage social risks, accelerate project approvals, and deliver complex infrastructure projects more efficiently. This capability can create a strategic advantage by improving project execution and reducing delays in highly regulated environments.

3. Strategic governance and investor confidence
Infrastructure projects involve complex regulatory frameworks, long investment horizons, and large capital commitments. As a result, strong governance structures and transparent decision-making processes are critical for managing project risks and maintaining investor confidence.

According to the OECD, **poor governance and mismanagement can account for 10–30% of infrastructure investment losses globally**, highlighting the importance of strong governance structures in large projects.[23]

Developers that embed strong ESG governance practices into project planning and oversight are better positioned to attract long-term institutional capital, navigate regulatory complexity, and secure approvals for large-scale projects. This capability can create a competitive advantage by improving project bankability, strengthening investor trust, and enabling faster execution of infrastructure investments.

Ørsted – ESG driving infrastructure transformation

Originally one of Europe's most coal-intensive utilities, Ørsted transformed its business model over the past decade by divesting fossil fuel assets and investing heavily in offshore wind infrastructure. As a result, the company reduced its carbon intensity by **over 87% between 2006 and 2022** while becoming one of the world's largest offshore wind developers.[24]

This strategic shift enabled Ørsted to secure government approvals, attract long-term institutional investors, and build strong stakeholder support for large-scale renewable infrastructure projects.

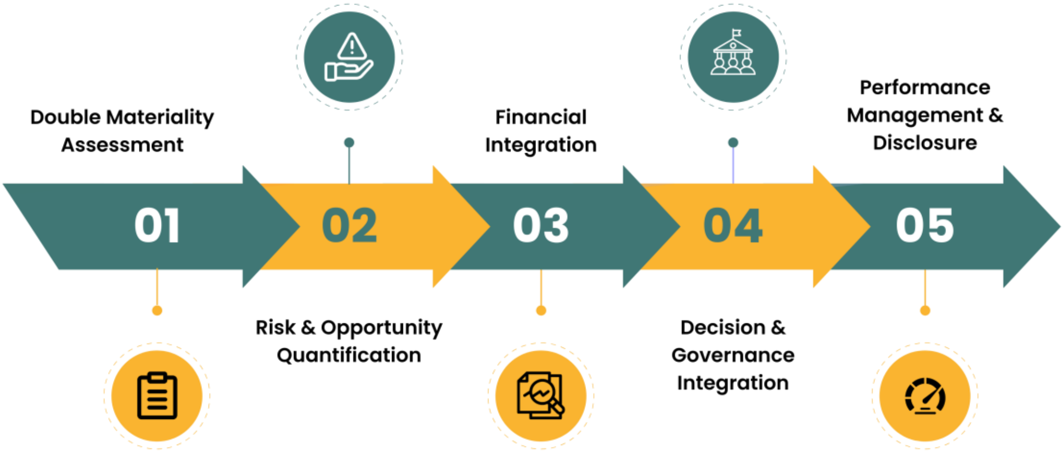
The company's transformation demonstrates how strong ESG performance can function as a **license to operate**, enabling infrastructure developers to access capital, navigate regulatory scrutiny, and deliver large-scale projects with greater certainty and investor confidence.

Across sectors, the mechanism differs - but the principle is the same: ESG creates competitive advantage only when embedded into the decisions that drive business performance.

21. Global Status Report for Buildings and Construction | UNEP - UN Environment Programme
22. Lessons from Four Decades of Infrastructure Project-Related Conflicts in Latin America and the Caribbean
23. GOV | OECD
24. orsted-esg-performance-report-2022.pdf

Integrating ESG into Strategic Business Decisions

The Five-Step Strategic ESG Integration Framework



Step 1 – Double materiality assessment

A double materiality assessment helps to clearly identify what matters to business and stakeholders. These include:

- **Financial materiality:** Issues affecting **revenue, costs, risk exposure, and access to capital**
- **Impact materiality:** Environmental and social impacts **across operations and the value chain:**

Without understanding material issues, ESG efforts scatter across low-impact initiatives while missing strategic opportunities.

Organizations typically follow a four-step process to conduct a comprehensive DMA:

Step	Action	Outcome
A. Stakeholder Engagement	Survey, interview or hold workshops with investors, customers, employees, suppliers, regulators	Understand external expectations
B. Impact Analysis	Assess environmental/social footprint of business across value chain	Identify where you create impact
C. Financial Linkage	Map ESG issues to revenue, costs, risks, capital	Prioritize business-critical issues
D. Materiality Matrix	Plot issues by financial + impact materiality	Focus areas for action

Step 2 – Risk & opportunity quantification

Once material sustainability issues are identified, the next step is to translate them into financial terms. Sustainability topics are often discussed qualitatively, such as climate risk, supply chain disruption, or regulatory pressure, but strategic decisions require measurable financial implications.

Quantification converts these issues into specific costs, savings, or revenue opportunities. This allows sustainability initiatives to be assessed alongside traditional business investments in capital allocation, strategic planning, and risk management processes. Management teams and boards can then compare options and prioritize actions that deliver both environmental and financial value.

Common approaches to quantify ESG risks and opportunities include:

Risk Type	Quantification Method	Example
Regulatory	Carbon pricing scenarios	CBAM: €75/tCO_2 x 100,000 tons exported = $7.5M annual cost
Physical	Climate scenario modelling	Increase in water costs in water stressed areas
Reputational	Revenue-at-risk analysis	ESG screening failure: 20% of buyers exclude supplier

Opportunity	Quantification Method	Example
Cost savings	Energy or waste reduction ROI	Renewable energy adoption leading to energy cost reduction
Revenue growth	Market sizing of sustainable products	Green product line give a price premium and volume growth
Capital access	Financing cost improvements	ESG-linked loan produce less interest

Maersk – Quantifying risks

Maersk, the world's largest container shipping company, faces significant exposure to future carbon pricing regulations. To prepare for this risk, the company integrated carbon pricing into its capital allocation framework, ensuring climate policy risk is reflected in investment decisions before regulations are finalized.

Carbon pricing to estimate future regulatory costs: Maersk applies an internal carbon price of **USD 75 per tonne** of greenhouse gas emissions when **evaluating new investments**. By assigning a monetary cost to emissions, the company can estimate the potential financial impact of future carbon regulation on projects and operations, ensuring climate policy risk is reflected in capital allocation decisions.[28]

Carbon pricing scenarios used to estimate regulatory risk: Industry modelling referenced by Maersk indicates that carbon prices could reach **up to $600/t$CO_2$** to incentivize low-carbon fuels in shipping, illustrating the potential regulatory cost exposure facing the sector.[29]

27. Dalmia-Bharat-IR-FY-24-25.pdf

Step 3 – Financial integration

Financial integration ensures that ESG considerations are not evaluated separately from business investments but are embedded directly into the criteria, models, and approval processes that shape where capital is deployed. This enables organizations to make trade-offs between sustainability objectives and financial returns within a unified framework, rather than treating ESG as a parallel or competing priority.

Integration Point	What it means	How to implement
CapEx evaluation	Add ESG criteria to capital project approval process	• Include carbon intensity per $ invested • Evaluate regulatory risk (e.g., CBAM exposure) • Assess resource circularity potential
OpEx optimization	Reframe sustainability as operational efficiency	• Include energy intensity targets in plant budgets • Apply internal carbon costs to business units • Track waste and water efficiency in operational KPIs
IRR Adjustment	Include ESG factors in risk-adjusted returns	• Include carbon pricing in cash-flow projections • Adjust discount rates for transition risk • Factor ESG-linked financing benefits
Capital allocation framework	Prioritize ESG investments at the portfolio level	• Create ESG investment criteria alongside financial hurdle rates • Require ESG impact disclosure in board-level capital reviews • Allocate capital pools for decarbonization initiatives • Create cross-functional investment committees

ONGC – Capital allocation for decarbonization

As global climate commitments and India's net-zero ambitions reshape the energy sector, ONGC developed a **decarbonization roadmap** to reduce its operational emissions and align with emerging climate expectations.

The roadmap outlines investments across renewable energy, hydrogen, CCUS, and electrification to achieve net-zero operational emissions by 2038. It also signals significant capital commitment, with ONGC planning to invest approximately ₹2 lakh crore across decarbonization initiatives over this period.

The plan includes large-scale capital deployment, such as ~**₹48,810** crore for renewable energy projects and evaluates emissions reduction potential for each intervention. [30]

However, financial integration alone is not sufficient. Sustaining ESG integration requires governance structures, clear ownership, and executive accountability. Step 4 focuses on embedding ESG into decision-making through board oversight, management incentives, and performance management systems.

28. 75fac9de-c2cc-421c-8eaf-17d5c5d86135
29. Maersk calls for $600 per tonne carbon price | myKN
30. ONGC Decarbonization Roadmap - en - ongcindia.com

Step 4 – Decision & governance integration

Organizations embed ESG into governance by assigning **board oversight, linking executive incentives to sustainability performance**, and **integrating ESG risks into enterprise risk management** frameworks. These mechanisms ensure that sustainability considerations remain embedded in strategic decision-making and operational accountability.

Mechanisms for ESG Governance Integration
1. Board-level Oversight

Governance Element	Implementation	Example
Board committee oversight	Assign ESG oversight to Sustainability Committee or Audit/Risk Committee	Board reviews climate transition plan annually
Board ESG KPIs	Track measurable sustainability targets at board level	Net-zero milestones, emissions intensity
Strategic review	Require ESG impact analysis for major investments	Board reviews emissions impact of new plant projects

2. Executive Compensation Integration
Companies increasingly link executive compensation to ESG performance to align leadership incentives with sustainability objectives, signal ESG priorities to investors and other stakeholders, and ensure management accountability for delivering on climate and social commitments.

In many Indian companies, ESG-linked metrics now account for roughly **10–15%** of executive performance KPIs, reflecting the growing integration of sustainability targets into leadership incentives.[32]

Companies typically integrate ESG into executive compensation, mechanisms such as:

- Link **10–20%** of variable executive pay to ESG KPIs
- Include ESG metrics in annual bonuses or long-term incentive plans
- Use **quantitative targets** such as emissions reduction or diversity goals
- Apply **scorecard or payout modifiers** tied to ESG performance

ESG KPIs are mapped to leadership roles based on functional responsibility:

Role	Example KPI
CEO	Scope 1 and 2 emission reduction percentage
CFO	Sustainable financing secured
COO	Renewable energy adoption percentage
Procurment Head	Supplier ESG compliance percentage

3. Enterprise Risk Management Integration
Companies integrate ESG considerations into Enterprise Risk Management (ERM) by incorporating climate-related risks into existing risk categories such as operational, regulatory, financial, and reputational risk.

This includes assessing physical climate risks to operations and supply chains, modeling exposure to carbon pricing and environmental regulations, evaluating impacts on financing conditions and cost of capital, and monitoring reputational risks associated with stakeholder expectations and ESG ratings.

32. Performance on sustainability targets shapes top execs' pay - The Economic Times

Tech Mahindra – Linking executive compensation

Tech Mahindra, a global IT services company headquartered in India, integrates ESG priorities into leadership performance through executive **Balanced Scorecards (BSC).** Material ESG topics such as **climate action, talent management, and customer satisfaction** are translated into measurable targets that influence the variable pay of senior leaders, including the CEO and other CXOs.

For example, metrics such as **GHG emissions reduction, employee engagement, gender diversity in leadership, and Net Promoter Score (NPS)** are incorporated into executive performance evaluations. This approach strengthens accountability by aligning leadership incentives with sustainability outcomes and long-term stakeholder value creation.[33]

Step 5 – Performance Management & Disclosure

Once ESG is embedded in investment decisions and governance, organizations must **measure performance, track progress, and communicate outcomes to investors and stakeholders**. Performance management ensures ESG initiatives deliver measurable business value and remain aligned with strategic targets.

1. Track ESG performance against strategic targets
ESG performance is monitored through KPIs directly linked to business objectives.

Strategic objective	ESG KPI	Target	Tracking
Cost efficiency	Energy intensity	-20% by 2030	Quarterly
Revenue growth	Revenue from sustainable products	50% by 2028	Annual
Supply chain risk	Scope 3 supplier coverage	80% by 2026	Semi-annual
Capital access	ESG rating improvement	A or better	Annual

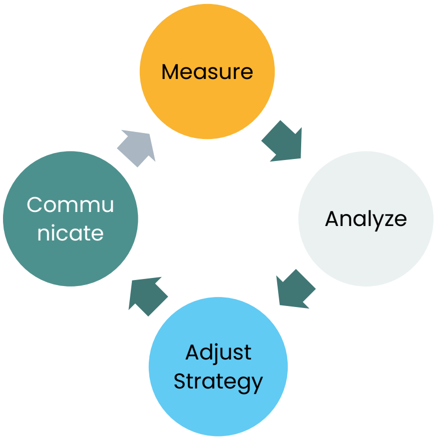
2. Integrate ESG into Corporate Reporting
Embed ESG performance into mainstream corporate disclosures rather than standalone sustainability reports. Examples include annual reports, investor presentations, quarterly earnings disclosures etc. This helps to signal investors that ESG performance is a core business driver.

3. Disclose using recognized frameworks
To ensure credibility and comparability, companies disclose ESG performance using standardized reporting frameworks.

33. [ESG Data Book 2024-25]

Category	Standards/Ratings
Regulatory frameworks	CSRD (EU), BRSR (India)
Global standards	TCFD, ISSB, GRI, SASB
Ratings and benchmarking	MSCI, Sustainalytics, CDP

4. Establish continuous feedback loops
Effective ESG performance management requires continuous monitoring and improvement. The typical cycle is shown below:



Key enablers include:

- Centralized ESG data platforms
- Automated operational data collection
- Executive dashboards for performance monitoring
- Audit-ready documentation for regulators and investors

Schneider sustainability impact framework

Schneider Electric, tracks its sustainability performance through the **Schneider Sustainability Impact (SSI)** framework. The program translates ESG commitments into measurable indicators across climate action, resource efficiency, social impact, and governance.

The SSI aggregates multiple sustainability KPIs into a **single score on a 10-point scale,** enabling the company to track progress against defined targets and communicate results transparently. In 2025, Schneider Electric reported an **SSI score of 8.88/10**, reflecting strong progress toward its sustainability goals.

Performance metrics span climate, social, and governance areas. For example, Schneider Electric's solutions have helped customers save and avoid **862 million tonnes** of CO_2 emissions, surpassing its 2025 ambition of 800 million tonnes. Importantly, SSI performance directly influences **short-term incentive compensation for executives and approximately 80,000 employees,** embedding sustainability accountability across the organization.[34]

34. Schneider Electric extra-financial results Q4 2025

From Ambition to Advantage

The organizations that will define the next decade are not those that report best on sustainability. They are those that use it to make better decisions. ESG is not a constraint on performance. It is a catalyst for it.

ECOPRISM

Where Strategy Meets System

ecoPRISM helps organizations embed ESG into the decisions that drive business value - from capital allocation and operational planning to performance tracking and investor disclosure.

Ready to move from ESG ambition to strategic execution?

Contact us